

CSCE 671 HUMAN-COMPUTER INTERACTION · MODULE 07 · LECTURE NOTES

Usability Testing

The method that makes this subject empirical.

MODULE OUTCOMES

- Run a think-aloud usability test.
- Explain why five participants is the right number.
- Write tasks that do not give away the answer.
- Avoid the ways a facilitator corrupts a session.
- Analyse a session into mechanisms and severities.

1 The method

1. WRITE 3-5 TASKS a real user would want to do, phrased as goals and not as instructions (section 3).
 2. SIT SOMEBODY DOWN and ask them to attempt the first task, thinking aloud as they go.
 3. DO NOT HELP. This is the hard part and it is the whole method.
 4. WATCH AND NOTE: where they hesitate, what they try, what they expected to happen, what they say, and whether they succeed or give up.
 5. REPEAT with four more people.
 6. ANALYSE: every failure attributed to a mechanism (Module 01 section 1) and rated by severity.
- THAT IS ALL. It takes a day, needs no equipment, and it is the only reliable source of information in this course (Module 01 section 2).

NOT HELPING IS THE WHOLE METHOD, AND IT IS GENUINELY DIFFICULT

Watching somebody struggle with something you built is uncomfortable — and **every hint you give destroys the data you came for**, because after the hint you are observing a person who has been told the answer.

So the facilitator's responses are scripted in advance: 'what do you think you should do?', 'what did you expect to happen?', and silence — **which is the most useful of the three** and the hardest to maintain.

And the participant has to be told that it is not a test of them, repeatedly and at the start of each task, because **they will apologise for your design's failures** — and **that discomfort suppresses what they say**, which is the material you are there for.

Which means the facilitator's job is to be boring and patient — and **the commonest failure of a first study is a facilitator who could not stop helping**, which produces a pleasant session and no findings.

2 Five participants

FIVE PARTICIPANTS FIND MOST OF THE PROBLEMS, BECAUSE THE PROBLEMS ARE NOT INDEPENDENT

If each participant has roughly a one-in-three chance of encountering any given problem, then five participants find a large majority of the problems present — which is the standard argument, rests on an assumed detection rate, and is approximately right in practice.

And the returns diminish sharply: the sixth participant mostly reproduces what the first five already showed — so **the marginal information is low and the marginal cost is not**, which is the actual argument for stopping.

So the efficient strategy is many small studies: five people, fix what you found, five more — rather than one study of twenty, which finds the same problems five times and delays every fix until the end.

But the argument fails for genuinely distinct user populations: five participants from one group does not cover another — so it is **five per population**, which is CSCE 632's point and is where the number grows legitimately rather than by excessive caution.

- **Distinct populations** — novices and experts, or people with different accessibility needs, **encounter entirely different problems** (CSCE 632) — so sampling one tells you nothing about the other.
- **Rare but severe problems**, which **five people will probably miss** — and which **heuristic review** (Module 05 §4) **may catch instead**, which is one of the better arguments for running both methods.
- **Any quantitative claim. Five participants cannot support a claim about a rate, a mean, or a difference**, which is **Module 08's territory** and requires a different design and a larger sample.
- **And comparison between two designs**, which needs either more participants or a within-subject design with counterbalancing (Module 08 §2).
- **So five is right for finding problems and wrong for measuring anything** — **which is the distinction that keeps this method honest** and is exactly what Module 13 asks you to state when reporting it.

3 Tasks

- **Phrase the task as a goal, not an instruction.** 'Find out how much you have spent this month' tests the design; 'click the Reports tab and then select Monthly' tests nothing at all except whether they can follow instructions.
- **Avoid the interface's own vocabulary**, because **using the label gives away the answer** — which is the **single commonest way a task is ruined**: the participant then searches for the word rather than for the function, and you learn nothing about whether the word was findable.
- **Make it realistic and specific, with the context a real user would have** — a task with **no motivation produces aimless behaviour**, and a participant who does not care about the outcome gives up differently than a real user would.
- **Ensure it is actually completable**, and **know the intended path before you start** — discovering mid-session that the task cannot be done wastes the participant and the slot.
- **And order the tasks so that an early one does not teach the later ones**, or **accept that it will and account for it in the analysis** — which is Module 08 §2's ordering effect appearing in a qualitative method.

4 Analysis

ATTRIBUTE EVERY FAILURE TO A MECHANISM, THEN RATE IT BY FREQUENCY TIMES IMPACT

A list of things that went wrong is not actionable. A list attributed to Module 01 §1's six mechanisms is, because each mechanism has a known family of fixes — which is why that vocabulary was introduced in the first module and why Project 1 grades the attribution.

And rate severity as frequency times impact: a problem that four of five people hit and that stops the task outranks one person's cosmetic complaint, however articulately expressed — and articulate complaints are over-weighted systematically.

Beware of over-weighting what participants *said* — what they did is the data and what they said is a hypothesis about it, which is Module 09 §3's distinction and the single most important analytical discipline here.

And report the things you expected to fail and which did not — because a study that confirmed everything you already believed was probably run badly, or analysed selectively (Project 1's fifth requirement).

- **Test early, when changing things is still cheap** — a study on a finished product finds problems nobody will fix, which wastes the study and discredits the method.
- **Have the people who will do the fixing watch the sessions**, because **watching one person struggle is more persuasive than any written report** of the same session — and the persuasion is what gets the fix scheduled.

- **Keep the studies small and frequent**, which is §2's argument and makes the method sustainable rather than an event.
- **Fix something before the next round**, so that the studies compound rather than each one producing an independent list.
- **And write down what you looked for and did not find, which is what makes the claim checkable** (Module 13 §2, and CSCE 713 Module 11 §4's recording requirement). **Having the implementers watch is the intervention that changes organisations — a written finding is arguable and a recording of somebody failing is not.**

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Krug — Rocket Surgery Made Easy	The whole module, practically — the cheapest possible version of the method, argued for convincingly. Library copy.
Nielsen — Why You Only Need to Test with 5 Users (free)	§2's argument in its canonical form — read it with the critiques, which are also linked there.
Dumas & Redish — A Practical Guide to Usability Testing	§1, §3, and §4 thoroughly — the task writing and the facilitation in detail. Library copy.
Nielsen Norman Group on think-aloud protocols (free)	§1's protocol , with the facilitator scripts and the common mistakes.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Write five tasks** for an interface, phrased as goals.
- 2 **Check each for the interface's own vocabulary** and rewrite any that leak it.
- 3 **Run one session** and count how many times you wanted to help.
- 4 **Run it again without helping** and compare what you learned.
- 5 **Run five sessions** and record every failure.
- 6 **Count how many new problems each participant added.**
- 7 **Attribute every failure** to one of the six mechanisms.
- 8 **Rate them by frequency times impact** and pick the top three.
- 9 **Separate what participants did from what they said**, and note where they disagreed.

- 10 **Have somebody who will fix it watch a session**, and record what changed afterwards.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Give the six steps of a usability test.
- 2 Why is not helping the whole method, and what are the three scripted responses?
- 3 Why does a participant apologise, and what does it cost you?
- 4 Give the five-participant argument and why returns diminish.
- 5 Name four cases where five is not enough.
- 6 What is five right for, and wrong for?
- 7 Give five rules for writing tasks.
- 8 What is the commonest way to ruin a study?
- 9 How do you analyse a session into actions?
- 10 Why report the things that did not fail?